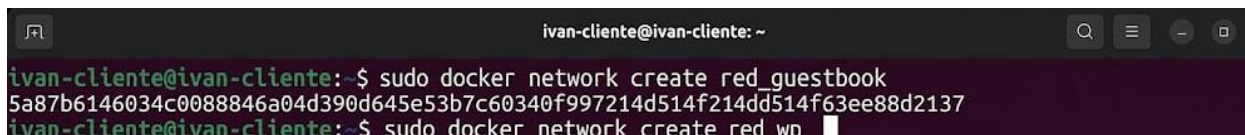


Actividad 4 - Almacenamiento y redes Docker

1. Despliegue de la aplicación Guestbook

Creemos la red interna que conectará los contenedores entre sí:

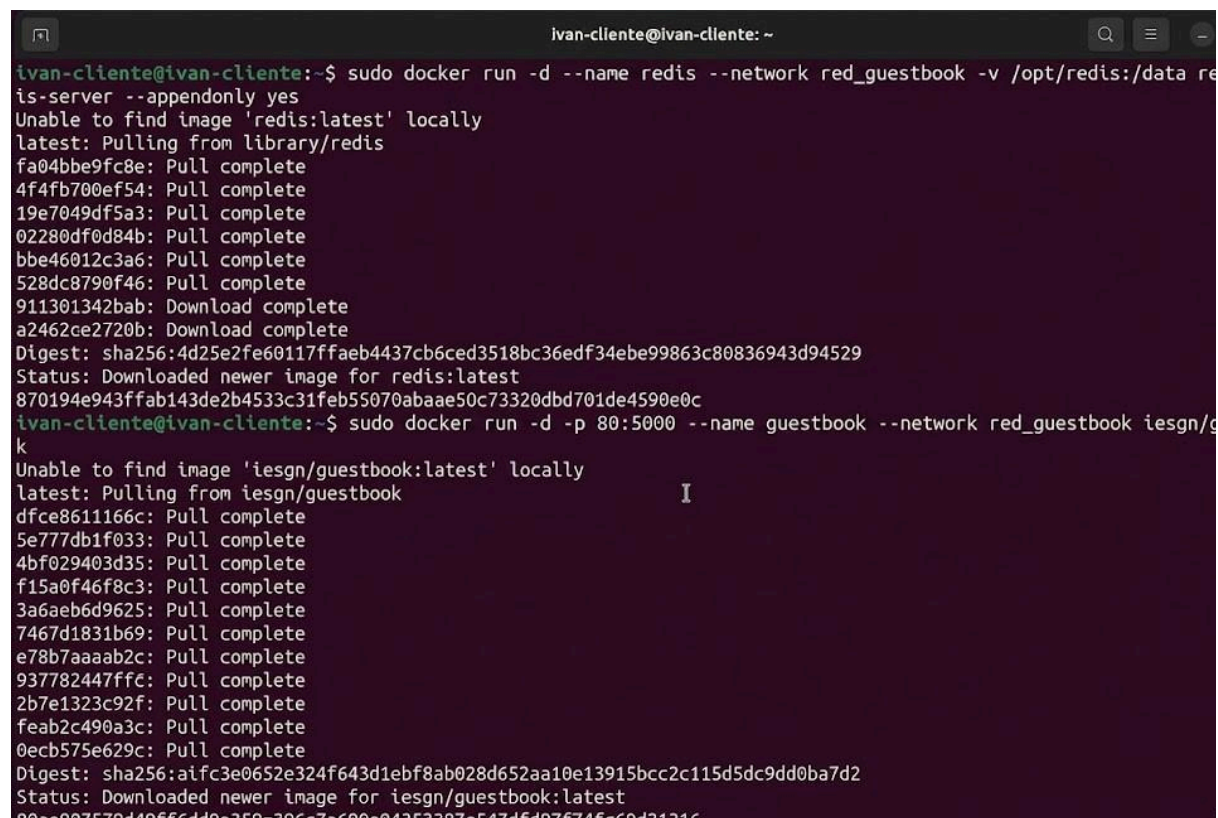
```
sudo docker network create red_guestbook
```



```
ivan-cliente@ivan-cliente: ~  
ivan-cliente@ivan-cliente:~$ sudo docker network create red_guestbook  
5a87b6146034c0088846a04d390d645e53b7c60340f997214d514f214dd514f63ee88d2137  
ivan-cliente@ivan-cliente:~$ sudo docker network create red_wp
```

Lanzamos el contenedor de Redis con un volumen persistente en /opt/redis y, a continuación, el contenedor de la aplicación Guestbook mapeando el puerto 5000 al 80:

```
sudo docker run -d --name redis --network red_guestbook -v /opt/redis:/data redis redis-server  
--appendonly yes  
sudo docker run -d -p 80:5000 --name guestbook --network red_guestbook iesgn/guestbook
```



```
ivan-cliente@ivan-cliente: ~  
ivan-cliente@ivan-cliente:~$ sudo docker run -d --name redis --network red_guestbook -v /opt/redis:/data re  
is-server --appendonly yes  
Unable to find image 'redis:latest' locally  
latest: Pulling from library/redis  
fa04bbe9fc8e: Pull complete  
4f4fb700ef54: Pull complete  
19e7049df5a3: Pull complete  
02280df0d84b: Pull complete  
bbe46012c3a6: Pull complete  
528dc8790f46: Pull complete  
911301342bab: Download complete  
a2462ce2720b: Download complete  
Digest: sha256:4d25e2fe60117ffaeb4437cb6ced3518bc36edf34ebe99863c80836943d94529  
Status: Downloaded newer image for redis:latest  
870194e943ffab143de2b4533c31feb55070abaae50c73320dbd701de4590e0c  
ivan-cliente@ivan-cliente:~$ sudo docker run -d -p 80:5000 --name guestbook --network red_guestbook iesgn/g  
k  
Unable to find image 'iesgn/guestbook:latest' locally  
latest: Pulling from iesgn/guestbook  
dfce8611166c: Pull complete  
5e777db1f033: Pull complete  
4bf029403d35: Pull complete  
f15a0f46f8c3: Pull complete  
3a6aeb6d9625: Pull complete  
7467d1831b69: Pull complete  
e78b7aaaab2c: Pull complete  
937782447ffc: Pull complete  
2b7e1323c92f: Pull complete  
feab2c490a3c: Pull complete  
0ecb575e629c: Pull complete  
Digest: sha256:aifc3e0652e324f643d1ebf8ab028d652aa10e13915bcc2c115d5dc9dd0ba7d2  
Status: Downloaded newer image for iesgn/guestbook:latest  
80e007579d49f56dd9e259e396c7e600e04352287e547d5d876745f69d21216
```

Comprobamos que la aplicación está funcionando accediendo a 127.0.0.1 en el navegador:



2. Despliegue de la aplicación Temperaturas

Creamos la red para que los dos contenedores puedan comunicarse internamente:

```
sudo docker network create red_temperaturas
```

Lanzamos el backend y el frontend conectados a la red, mapeando el puerto 3000 al 80:

```
sudo docker run -d --name temperaturas-backend --network red_temperaturas iesgn/temperaturas_backend
sudo docker run -d -p 80:3000 --name temperaturas-frontend --network red_temperaturas
iesgn/temperaturas_frontend
```

```
ivan-cliente@ivan-cliente: ~
ivan-cliente@ivan-cliente:~$ sudo docker run -d --name temperaturas-backend --network red_temperaturas iesgn/temperaturas_backend
Unable to find image 'iesgn/temperaturas_backend:latest' locally
latest: Pulling from iesgn/temperaturas_backend
7238ddc8ca51: Pull complete
4a604c2354e7: Pull complete
f70ec8148670: Pull complete
32de3c850997: Pull complete
Digest: sha256:ad95f9721e81ac004e258ff0241aa6417c58c5c4f32821e7ff06e92926aa0d3b
Status: Downloaded newer image for iesgn/temperaturas_backend:latest
e2d0a0dd2423ae28f2aa065212d01dcbb74e9359374d32694608048fccb3e1
ivan-cliente@ivan-cliente:~$ sudo docker run -d --name temperaturas-frontend --network red_temperaturas iesgn/temperaturas_frontend
Unable to find image 'iesgn/temperaturas_frontend:latest' locally
latest: Pulling from iesgn/temperaturas_frontend
d724eea63276: Pull complete
1b0ea791d129: Pull complete
18ffb3e704a1: Pull complete
Digest: sha256:7a2b7b66f830e0905d2b0b7faf33a5bea3da7c12c8ea75abdc034603307c7677
Status: Downloaded newer image for iesgn/temperaturas_frontend:latest
16858eae652b669a855fa4bc94591cfaaf54cb247e6a0050fa2fb171e407c102
ivan-cliente@ivan-cliente:~$
```

```
ivan-cliente@ivan-cliente: ~
ivan-cliente@ivan-cliente:~$ sudo docker run -d --name temperaturas-backend --network red_temperaturas iesgn/temperaturas_backend
9fc4b4b6d847b161791ef6c745e605edee826c7d2639ec66f77075bf126912
ivan-cliente@ivan-cliente:~$ sudo docker run -d -p 80:3000 --name temperaturas-frontend --network red_temperaturas iesgn/temperaturas_frontend
7f52c906877f802088562c8fd7a28886f46ee88bd912c778f9f8054cd0a1dd17
ivan-cliente@ivan-cliente:~$
```

Comprobamos que la aplicación funciona correctamente en el navegador:



Temperaturas

Introduce el municipio (indica el nombre completo o el inicio) de España que quieres buscar...

Curso de Kubernetes. 2021 - hostname: 7f52c906877f

3. Despliegue de WordPress + MariaDB

Creamos la red que usarán los dos servicios:

```
sudo docker network create red_wp
```

```
ivan-cliente@ivan-cliente: ~  
ivan-cliente@ivan-cliente:~$ sudo docker network create red_wp  
47219c6280195bfe9d50291987082323d2467ac565f6715ed9fd8e4e66c91a69  
ivan-cliente@ivan-cliente:~$
```

Lanzamos el contenedor de MariaDB con las variables de entorno necesarias y un volumen persistente en /opt/mysql_wp:

```
sudo docker run -d --name servidor_mysql --network red_wp \  
-v /opt/mysql_wp:/var/lib/mysql \  
-e MYSQL_DATABASE=bd_wp -e MYSQL_USER=user_wp \  
-e MYSQL_PASSWORD=prueba -e MYSQL_ROOT_PASSWORD=prueba mariadb
```

```
ivan-cliente@ivan-cliente:~$ sudo docker run -d --name servidor_mysql \  
>  
> .bash_history Descargas/ .local/ snap/  
> .bash_logout Documentos/ Música/ .ssh/  
> .bashrc Escritorio/ Plantillas/ .sudo_as_admin_successful  
> .cache/ hola.txt .profile Videos/  
> .clamtk/ Imágenes/ Público/  
> .config/ informe_clamav.txt ShellScripting/  
> --network red_wp \  
> -v /opt/mysql_wp:/var/lib/mysql \  
> -e MYSQL_DATABASE=bd_wp \  
> -e MYSQL_USER=user_wp \  
> -e MYSQL_PASSWORD=prueba \  
> -e MYSQL_ROOT_PASSWORD=prueba \  
> mariadb  
c1ef2a6ba760a22c0890e9548e3a132b9d6b86a77f4b5fca4811f9acc1d20fe7  
ivan-cliente@ivan-cliente:~$
```

Lanzamos el contenedor de WordPress apuntando a MariaDB como host de base de datos, con un volumen para el contenido y mapeando el puerto 80:

```
sudo docker run -d --name servidor_wp --network red_wp \  
-v /opt/wordpress:/var/www/html/wp-content \  
-e WORDPRESS_DB_HOST=servidor_mysql -e WORDPRESS_DB_USER=user_wp \  
-e WORDPRESS_DB_PASSWORD=prueba -e WORDPRESS_DB_NAME=bd_wp \  
-p 80:80 wordpress
```



```
ivan-cliente@ivan-cliente: ~  
ivan-cliente@ivan-cliente:~$ sudo docker run -d --name servidor_wp \  
> --network red_wp \  
> -v /opt/wordpress:/var/www/html/wp-content \  
> -e WORDPRESS_DB_HOST=servidor_mysql \  
> -e WORDPRESS_DB_USER=user_wp \  
> -e WORDPRESS_DB_PASSWORD=prueba \  
> -e WORDPRESS_DB_NAME=bd_wp \  
> -p 80:80 \  
> wordpress  
Unable to find image 'wordpress:latest' locally  
latest: Pulling from library/wordpress  
b81875e03a86: Pull complete  
3bcd9b27175b: Pull complete  
c7a376359321: Pull complete  
87574aead5d7: Pull complete  
1cf5064b55f1: Pull complete  
eca30975eae4: Pull complete  
c37670fe2f1e: Pull complete  
320d77ecabde: Pull complete  
4f4fb700ef54: Pull complete  
dc404e2c7d84: Pull complete  
66d57c3a38eb: Pull complete  
fa05b5032b13: Pull complete  
8233764c9d8b: Pull complete  
6bbcb046e4ae: Pull complete  
ba11284aac47: Pull complete  
40ec4bb877a9: Pull complete  
8f8f3a474090: Pull complete  
edfa1eac0ec9: Pull complete  
862ee81cb220: Pull complete  
886235a2a1ff: Pull complete  
0aa392ebbb80: Pull complete  
816a9fd16934: Pull complete
```

Accedemos a 127.0.0.1 en el navegador y comprobamos que aparece el instalador de WordPress:

